

HENRIQUE DERVICHE KASPRZAK

FULL-STACK DEVELOPER · GO & REACT

LOCATION

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SUMMARY

Full-stack developer and Computer Engineer with real production experience. I work on a smart-city platform, building Go microservices that integrate cameras, IoT sensors and urban systems in real time, along with React interfaces.

Outside work, I ship complete systems on my own — from a Go fleet-management system with an offline-first app and a cryptographic audit trail, to an IoT capstone with a Go API and a React dashboard. I value simple, direct solutions: a lean HTTP handler over a framework, and understanding the problem over adding a dependency. I use AI as a real productivity tool without giving up understanding what I build.

HIGHLIGHTS

- Production Go microservices: real-time data, webhooks, MQTT, observability and Kubernetes deployment.
- End-to-end full-stack systems delivered solo — backend, frontend, database, deployment, tests and docs.
- Software/hardware integration: IoT, sensors, ESP32/Raspberry Pi and real-time protocols.
- Quality focus: CI tests, database migrations, disaster recovery and ADRs.

TECHNICAL SKILLS

Languages

Go, JavaScript, TypeScript, C, C++, Java, SQL, Bash, HTML and CSS.

Front-end

React 19, Vite, Tailwind CSS, Recharts, Radix UI, Framer Motion, Leaflet and htmx.

DevOps & cloud

Docker, Docker Compose, Kubernetes, GitHub Actions (CI), Render, Vercel and MagaLU Cloud.

IoT, hardware & domains

ESP32, Raspberry Pi, ADC sensors, computer vision, LPR/ANPR, facial recognition, PTZ camera control and urban telemetry.

Back-end & real-time

REST APIs, WebSocket (gorilla/websocket), MQTT (Paho/Mosquitto), webhooks, Chi router, microservices and multi-tenancy.

Data & infrastructure

PostgreSQL/PostGIS, Redis, Supabase, MinIO/S3, migrations (Goose) and pgx.

Observability & practices

Prometheus, OpenTelemetry, structured logging (Zerolog), testing, ADRs and retry/backoff patterns.

Tooling & AI

Git/GitHub, CLI, advanced use of LLMs for development and code review, local models (LM Studio), GIMP and Inkscape.

PROFESSIONAL EXPERIENCE

Metropolys — Software Developer

Smart-city platform | Oct 2024 – Present | Curitiba, Brazil

- Built, as primary author, a Go License Plate Recognition (LPR/ANPR) webhook integration service: plate-image download and upload, vehicle-metadata extraction, restriction-list checks with priority-based alerting and SHA-256 event deduplication, exposing Prometheus metrics.
- Authored the Go service for PTZ (pan-tilt-zoom) camera control and its React 19 overlay frontend — directional dial, zoom/focus/iris controls and custom command hooks rendered over the live video stream.
- Created several Go urban-telemetry simulators (traffic lights with failure modes, public lighting, GeoJSON route-based GPS tracking and facial detection), enabling end-to-end pipeline testing without physical hardware, using goroutines and mutexes.

- Developed a KPI/indicator aggregation service that combines traffic, weather and urban-infrastructure data into standardized multi-tenant metrics.
- Contributed to integration services for real-time public transit, IoT sensor telemetry and public lighting over MQTT.
- Built a React camera-mosaic component (draggable, resizable tiles, saved presets, search and a Leaflet map) integrated into the operations dashboard, plus per-city Kubernetes deployment.

Go PostgreSQL/PostGIS Redis MQTT WebSocket Docker Kubernetes React Prometheus

FEATURED PROJECTS

Xanadu Fleet — Fleet inspection & maintenance management system

Complete system built end-to-end as sole author (~130 commits)

- Operational fleet inspections through versioned digital checklists (58 truck items, 19 machine items) executed at vehicle hand-out and hand-back, with immutable version pinning.
- Offline-first PWA for field crews to capture inspections and photos without connectivity, with later sync; an html/template + htmx web admin for assets, users, roles and reports.
- Role-based access control (driver, supervisor, admin), audit trails, nonconformity workflows and a cryptographic evidence chain (WORM envelope) guaranteeing record integrity.
- Database migrations (Goose), unit and integration tests in CI, production hardening (upload timeouts, graceful shutdown) and disaster recovery (pg_dump backups + PITR), documented with 7 Architecture Decision Records.

Go PostgreSQL htmx Offline-first PWA MinIO/S3 GitHub Actions MagaLU Cloud

Mini Estufa — Capstone · Real-time full-stack IoT

Greenhouse monitoring integrating hardware, sensors and a web app

- Go API (sole author) receiving telemetry from the hardware (ESP32) over HTTP and broadcasting it in real time over WebSocket, with /api/sensor/push, /api/sensor/latest, /ws and /health handlers.
- React 19 + Vite + Tailwind dashboard with Recharts graphs (temperature, air humidity, light level, soil moisture and pump/light status) and history persisted in Supabase, with date-range filtering and timezone handling.
- Physical assembly, ADC sensor reading and calibration, irrigation-pump automation and integration with an MQTT/Mosquitto pipeline; API deployed on Render and dashboard on Vercel.

Go gorilla/websocket React 19 Vite Tailwind Recharts Supabase MQTT Render Vercel

Xanadu Site — Institutional website

Responsive institutional website built as sole author

- Landing page with Framer Motion animated sections, a Cloudinary-hosted hero video, a contact form with WhatsApp integration and a location map.
- Deployed on Vercel with preview-to-production domain redirects and SEO headers.

React 19 Vite Framer Motion Tailwind Cloudinary Vercel

EDUCATION, COURSES & LANGUAGES

Universidade Positivo

B.Sc. in Computer Engineering | 2021 – 2026

Portuguese: native.

English: reading, writing and conversation.

Go (Golang): Udemy, 60 hours

React with REST APIs: Udemy, 30 hours

GitHub: Udemy, 20 hours

Dale Carnegie: communication, leadership and interpersonal skills